

Project 2: Numerical solutions to incompressible Navier–Stokes equations

Important notes before you start:

1. Your project will be evaluated based on your submitted report. Present your work in the style of a journal article using scientific languages and support your discussions using appropriate figures. Please use the `aapm` class inside the `REVTeX 4.2e` package as the template of your report.
2. Your report to the project is **limited to 10 pages** using the required template (including references). You are required to append your code to your report, and this is not counted towards the 10-page limit. You can simply export/print your code into a single PDF and append it to your report and no need to include that in the `LATEX` code of your report.
3. Your analysis and exploration should go beyond what are being asked by the statement of your project.
4. Presentation really matters. For scientific papers and formal reports, good presentation is not a subjective matter. It usually comes with guidelines. You are encouraged to ask me for tips and feedback to improve the presentation of your document. The more your document looks like an actual scientific paper, the better grade you will get. Your interpretation and presentation of your results are emphasized as much as the results themselves.

Numerical simulations of lid-driven cavity flows

Numerically simulate the lid-driven cavity flows by solving the incompressible Navier–Stokes equations using the fractional-step method with 2nd-order Adam–Bashforth (AB2) scheme for the advection term and Crank–Nicolson (CN) for the viscous term [1]. Follow the problem settings reported in Ghia, Ghia, and Shin [2] and validate your results against those reported in the paper (e.g., tables 1, 2 and figures 3, 4, etc.). Validation needs to be performed for at least one Reynolds number. Explore interesting cases with your powerful incompressible Navier–Stokes solver that you code up from scratch!

References

- [1] J. B. Perot. An analysis of the fractional step method. *Journal of Computational Physics*, 108(1):51–58, 1993.
- [2] U. Ghia, K. N. Ghia, and C.T. Shin. High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method. *Journal of Computational Physics*, 48(3):387–411, 1982.